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Evolving Visual Odometry for Autonomous Underwater Vehicles ⋆

Bo Miquel Nordfeldt Fiol * **Francisco Bonin-Font** *
Gabriel Oliver Codina * **Yolanda Gonzalez Cid** *

* *Systems, Robotics and Vision Group, University of the Balearic Islands, ctra de Valldemossa km 7.5, Palma de Mallorca (Spain)*

Abstract: In the last years visual odometers have been improved considerably, but there is still a certain lack of robustness and reliability when used to navigate *Autonomous Underwater Vehicles* (AUV) in complex underwater scenarios, such as habitats colonized with Posidonia oceanica or Zostera noltii. The work presented in this paper goes one step beyond the current solutions, improving a state of the art approach to be applied in underwater applications. Tests conducted with real marine visual data grabbed in waters of the Balearic Islands from a bottom-looking camera installed on an AUV, show the progress in the vehicle displacement estimation and its viability to be used online.

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1. INTRODUCTION

Autonomous Underwater Vehicles (AUV) usually base their navigation abilities in a set of sensors such as *Inertial Measurement Units* (IMU) to get orientation, acceleration and angular velocity, acoustic *Short baselines* (SBL or USBL) to get global localization, a *Doppler Velocity Log* (DVL) to get lineal velocity and navigation altitude underwater, and optical cameras which permit to compute the vehicle displacement and speed in up to 6 *Degrees of Freedom* (DoF).

Odometry is understood as the relative robot motion estimated in a time interval between two consecutive poses, and it is usually expressed as a pose transform or a velocity. In underwater robots, odometry estimated with IMUs or magnetic sensors is prone to generate and accumulate important drifts and, although DVLs or ecosounders can mitigate this problem in short term motion estimations, they cannot solve it in mid/long term missions. The addition of acoustic baselines offer absolute position with respect to a fixed point, but introduce interference in the trajectories, in the form of multiple echoes, outliers or saw tooth leading edges.

Cameras and visual odometry have been widely used to improve the motion estimation of robots (Nister et al., 2004; Kostavelis et al., 2016), since they have important advantages with respect to the other instruments: they provide real information that can be directly interpretable by humans, can record the color of the scene and have much higher temporal and spatial resolutions. Visual odometry can be defined as the process of incrementally estimating the pose of the vehicle by examining the changes that motion induces on the images of its onboard cameras (Scaramuzza and Fraundorfer, 2011). It is important to note that the definition includes the concept of pose estimation, rather than, motion estimation, implying the former the integration of the successive displacements obtained frame to frame, and consequently, accumulating drift. Neural networks have also been introduced to improve visual odometry in terrestrial and aerial autonomous vehicles, generating algorithms that do not require the camera calibration to estimate its movement (McConville et al., 2020; Gadipudi et al., 2021).

Visual odometers are usually constituted by two main blocks: 1) a feature tracker, and 2) a motion estimator. Underwater scenarios pose a lot of inconveniences to the imaging systems, such as, light reflection, refraction, absorption, haze, vigneting, blur, lack of contrast, and, more importantly, complex or intricate textures on the sea bottom that convert feature detection and matching into a very challenging task. Many researchers have addressed this later problem proposing specific solutions to enhance the quality of underwater images as a first step to increase the precision of visual odometry (Hu et al., 2022). Some other authors have focused on improving the distribution of visual features for underwater imagery turning to bucketing techniques (Negre-Carrasco et al., 2016; More et al., 2017), in order to avoid that most of the keypoints accumulate in the moving part of the image and the restrictions it causes in the motion estimation. Several improvements in stereo odometers for AUV are however based on visual features non invariant to changes on scale, rotation and/or translation (Nawaf et al., 2018; Dabove et al., 2020), tested with cameras pointing to the forward direction with a pitch of 45°, in short or mostly rectilinear transects, with few rotations, even with an insufficient number of field tests or performed only with publicly available underwater benchmarks (Dabove et al., 2020). None of these solutions assumed that underwater

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seabeds can present dynamic or complex textures, and that imagery might need to be grabbed with the cameras pointing perpendicularly to the bottom, while moving at a constant altitude. This latter configuration reduces the complexity of the visual odometry calculation to a 3 DoF (x,y, θ), where x and y represent the vehicle motion on the horizontal plane and θ its rotation in yaw.

In the context of the Spanish projects DETECPOS and COOPERAMOS, an AUV model SPARUS II (Carreras et al., 2018) property of the *University of the Balearic Islands* (UIB) needs to navigate across marine habitats between 4 to 40 meters deep, all colonized with extended seagrass meadows. The control architecture of the AUV is ROS-based (Stanford Artificial Intelligence Laboratory et al., 2022), and it integrates the navigation data provided by a DVL, a pressure sensor, an IMU, an ecosounder, and an USBL, to estimate the vehicle global pose and velocities (*localization data*) (Guerrero-Font et al., 2017). However, data coming from the DVL tends to vanish when the AUV navigates just over and close enough to the seagrass, causing frequent discontinuities of velocity and altitude data that can hamper a smooth vehicle navigation. For this reason, adding new redundant sensors to the navigation module is necessary, and, since the vehicle is also equipped with a stereo rig pointing downwards, a visual odometer is a good option. Previous work already incorporated the well known stereo odometer LIBVISO2 (Geiger et al., 2011) in AUVs, configuring it properly to get excellent results (Negre-Carrasco et al., 2016). LIBVISO2 outdoes other visual odometers in the way it eliminates outliers in visual correspondences applying a circular matching.

This paper presents an evolution of the LIBVISO2 stereo odometer, to be used in underwater scenarios colonized with different types of seagrass (Posidonia oceanica) or algae. This improvement responds to several unresolved issues: 1) the need of strengthen the visual feature tracking process in underwater challenging scenarios, 2) its application online, during observation and data collection missions, and c) the possibility of having slight motion on the seagrass observed on the seabed. In our case, 4 important considerations need to be taken into account: a) given the slow dynamics of the vehicle for these kind of campaigns and the low frame rates applied to grab video sequences, robustness will be preferable than speed, b) cameras will be always pointing downwards, with its lens axis perpendicular to the AUV horizontal axis, c) the stereo version of LIBVISO2 was chosen because, first, our vehicle will move in surge, heave and yaw, second, previous experimental tests (Wirth et al., 2013) showed a better performance of the stereo with respect to the mono-odometer, and third, the source code developed must not be effective only for our particular vehicle but be prepared for any other AUV with up to 6 DoF; finally, d) last but not least, the objective is not using the data given by the visual odometer alone, but as an additional source of information to be integrated in the navigation module of the AUV, in case any of the other navigation instruments could fail.

2. THE LIBVISO2 STEREO ODOMETER

The main steps of the LIBVISO2 stereo odometer are (Geiger et al., 2011):

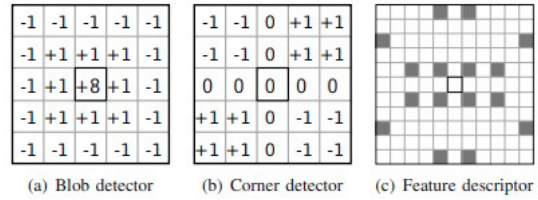

Fig. 1. (a) Blob and (b) corner detectors. (c) A possible descriptor with the 16 highlighted cells that contain the Sobel filter responses (Geiger et al., 2011).

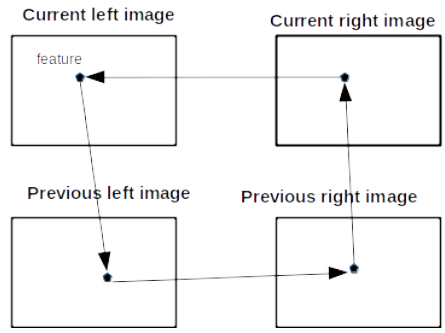

Fig. 2. Ilustration of the circular matching.

1) Visual features are extracted in consecutive stereo pairs filtering both, left and right images, with the 5 × 5 blob and corner masks of Figures 1-(a) and 1-(b), respectively. Non-maxima and non-minima suppression techniques are applied to reduce the number of obtained visual points. This type of feature detection assumes smooth trajectories and no significant changes on scale, rotation or translation, something that does not fit with the kind of missions to be performed in our projects.

2) Descriptors are composed by 11×11 windows resulting from a Sobel filter. Matchings between descriptors are found using the *Sum of Absolute Differences* (SAD) error metric.

3) A circular matching process is applied separately for each key point: each feature of the *current left image* is tracked in the *previous left image*. Next, the matched feature in the *previous left image* is tracked in the *previous right image*, and this match is tracked afterwards in the *current right image*. Finally, if the correspondence has survived, it is searched again in the *current left image*. If in the last matching, the feature pair of the *current right image* coincides (with a certain tolerance) with the original feature that started the matching in the *current left image*, the circular matching is considered valid. Matchings between left and right images include the requirement of the epipolar constraint with a tolerance of 1 pixel. The idea is illustrated in Figure 2

4) The remaining inliers are bucketed to homogenize their distribution over the image domain, causing a reduction of the definitive number of accepted circular matchings.

5) Finally, the algorithm computes the camera motion minimizing the sum of reprojection errors: 3D points obtained from features matched in a stereo pair are reprojected on the image domain again, causing a difference in location, minimized globally for all points under a RANSAC scheme.

Table 1. Properties of detectors

	Corner detector	Blob detector	Descriptor	Rotation invariance	Scale invariance	Repeatability	Localization accuracy	Robustness	Efficiency
Harris	x			x		+++	+++	++	++
Shi-Tomasi	x			x		+++	+++	++	++
FAST	x			x	x	++	++	+	++++
FREAK			x	x	x	++	++	++	++
SIFT		x	x	x	x	+++	++++	+++	+/2
SURF		x	x	x	x	+++	+++	++	+
ORB	x		x	x	x	+++	++	++	+++
BRISK	x		x	x	x	+++	+++	++	++

3. THE EVOLVED LIBVISO2 STEREO ODOMETER.

The objective is to get a visual odometer to complement the navigation data of an AUV that moves in underwater areas colonized with seagrass. The visual odometer has to be resilient to image rotations, changes on scale, challenging textures and lighting conditions because: 1) AUV missions will be mostly lawn-mower shaped, with several parallel linear transects but also many turns that cause important camera rotations, 2) vertical oscillations will also be usual due to the hysteresis of the vehicle altitude controller, 3) underwater environmental conditions can change, including the sea bottom or the clarity of water, among others. LIBVISO2 has been changed mostly in the feature matching block, leaving untouched the ego-motion estimation module, and converting the bucketing step into an optional element, selectable depending on the environment. According to (Bonin-Font et al., 2017), the best combinations of feature detectors and descriptors for environments with seagrass Posidonia oceanica are: a) SIFT detector combined with its own descriptor (SIFT-SIFT), b) SURF detector with the SIFT descriptor (SURF-SIFT), c) SURF detector with the BRISK descriptor (SURF-BRISK) and d) SURF detector with FREAK descriptor (SURF-FREAK). Table 1 (Forero et al., 2021) shows that all the components of these aforementioned combinations are robust in variable lighting conditions, invariant to rotations and scale changes (except FREAK, which this invariance is effective only if it is coupled with a detector providing it), and have good repeatability and localization accuracy. Crosses mark presence or absence of an ability and number of plus symbols mark the performance.

The improvements on LIBVISO2 consist in:

a) Feature detection has been changed to SIFT or SURF, because they are invariant to changes on scale, rotation, translation and illumination conditions, b) adopted descriptors include SIFT as unique floating point representative, or, alternatively, BRISK or FREAK as the binary; the FLANN-INDEX-KDTREE-SINGLE algorithm (Muja and Lowe, 2009) has been used with the L2-norm as a similarity metric for SIFT, while a brute-force technique with the Hamming-norm is used to compare binary descriptors, c) contrarily to the LIBVISO2 original version where the circular matching is forced feature by feature, circular matching is now computed globally, frame to frame, previously to the ego-motion calculation, in such a way that, outliers between left/right images are globally filtered using RANSAC and imposing the epipolar constraint and correspondences between current and previous frames are also jointly filtered with RANSAC and forced to satisfy a homography between them.

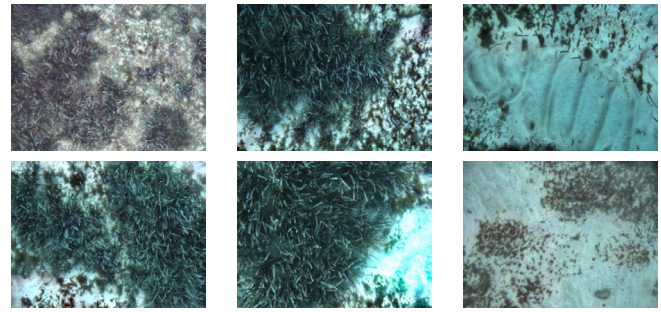

Fig. 3. Some images of the dataset grabbed in Portals Vells.

All these changes have been integrated in the LIBVISO2 ROS wrapper (Massot et al., 2013) using OpenCV functions, and made available for the research community.

4. EXPERIMENTAL RESULTS

4.1 Experimental setup

Preliminary experiments have been run with *bagfiles* grabbed from a stereo camera formed by two synchronized AVT Manta G-283, mounted on our Sparus II, pointing downwards, with their lens axis perpendicular to the longitudinal vehicle axis. *Bagfiles* are the main ROS mechanism for data logging, containing the AUV navigation topics and images, all time-stamped, registered and captured by the robot in a perfectly synchronous way. ROS permits to play them back, which generates the same data flow as having the same ROS nodes sending the same data with the same timestamps as all was running in real time, during the mission. Datasets were taken during several campaigns along the south of the Balearic Coast. The AUV was programmed to navigate at a constant altitude in all cases, but, due to the tolerance on the height controller, it usually fluctuated up and down between 0.5 and 0.7 meters around the set navigation altitude. Images were taken at 2 fps, with an original resolution of 1920×1440 pixels, debayerized, rectified and finally decimated by 4. The seafloor appeared to be quite heterogeneous with long leaves of Posidonia oceanica, pebbles, algae, low and complex textured sandy areas, and, different intensities of flickering.

A group of experiments used to illustrate this paper was performed in *Portals Vells* (bay of Palma). One sample survey was composed of 9 transects with a length of 38 m each one, separated 0.6 m. The AUV moved during 34 minutes, at a constant speed of 0.25 m/s, and at an approximate altitude of 2 m. This survey includes 3919 images. Figure 3 shows some images extracted from this dataset. Bagfiles were reproduced with a labtop Intel i7-6500U, at 2.50 GHz, with 2 cores (4 threads). A ROS node wrapping the original and the evolved LIBVISO2 were run in the

same ROS core as the tested *bagfiles*, but the improved version was configured with 8 different combinations of feature detectors and descriptors, as suggested in (Bonin-Font et al., 2017): SIFT/SIFT, SIFT/SIFT-No Bucketing, SURF/SIFT, SURF/SIFT-No Bucketing, SURF/BRISK, SURF/BRISK-No Bucketing, SURF/FREAK and with SURF/FREAK-No Bucketing.

Finally, concerning the operating parameters: a) In the original LIBVISO2, *nms-n* and *nms-tau* were set to 3, *matching binsize*=50 pixels, *matching radius*=200 pixels, *matching disparity tolerance*=1 pixel, *multi-stage*= true and *refinement*=1, *outlier disparity tolerance* and *outlier flow tolerance*= 5 pixels, b) in the new version that uses SIFT as a feature detector *Edge Threshold*= 10, *Contrast Threshold*=0.03, *standard deviation sigma*=1.6 c) in the new version with SURF *number of Octaves*=4, *hessian Threshold*=500, d) for both, SIFT and SURF, *number of Octaves per Layers*=3, *homography reprojection Threshold*=3 pixels and *epipolar constrains*= 1 pixel, and, e) for all combinations: *max number of features per bucket*=2, *bucket-width*= *bucket-height* =50 pixels, *RANSAC iterations*=200.

4.2 Results

All versions of LIBVISO2 output an *Odomery* ROS message containing the successive global vehicle poses, and the lineal and angular velocities, with their respective covariances. The library also provides the execution time per circular matching and motion calculation, the number of matches after bucketing and the number of final inliers after the motion estimation. In order to assess the performance of the visual odometers, the AUV position, orientation and lineal velocity computed continuously by all configurations of LIBVISO2 described in section 4.1 were compared with the AUV *localization data* estimated as referred in section 1 and published during the reproduction of the *bagfiles*. The difference between both data sources, the visual odometry and the original AUV *localization data*, defined the trajectory error, either point to point or in terms of a mean value. The AUV global *localization data* was assumed as the *Ground Truth* baseline since it was the most reliable and available measurement according to (Guerrero-Font et al., 2017). The mean of these errors are also presented for the sake of simplicity and summarization. Tables 2 and 3 show some quantitative results in the form of mean errors in pose and velocities, running times and matchings, for the aforementioned sample mission. Take into consideration that the mean pose error will be presumably high, for all cases, since it is calculated at the end of the trajectory, and can be also seen as a measure of the final accumulated drift. Although in dead-reckoning or odometry, comparing global pose data is not fair, figures 4, 5, 6 and 7 show the trajectories estimated by the different configurations of LIBVISO2, for the sake of a first qualitative evaluation.

Concerning the data shown in Table 2: a) the mean error in angular velocity in LIBVISO2, LIBVISO2-SURF-SIFT and LIBVISO2-SURF-SIFT-NoBuck is slightly higher than the other versions of the odometer, b) the highest pose (position and yaw) mean errors appear for the original LIBVISO2 and LIBVISO2-SURF-SIFT, with and without bucketing, c) the highest lineal velocity mean errors

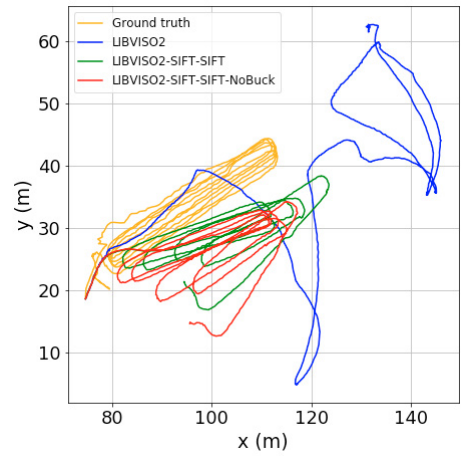

Fig. 4. Trajectories estimated by the original LIBVISO2 and LIBVISO2-SIFT-SIFT, vs the Ground Truth

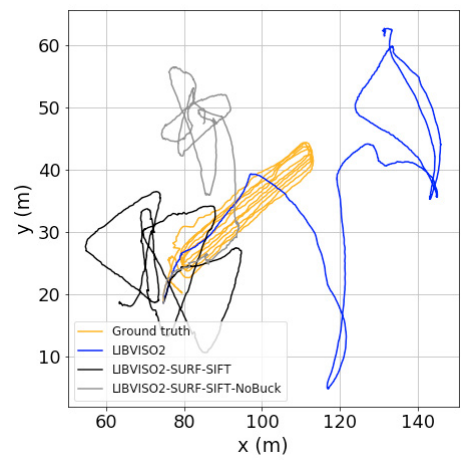

Fig. 5. Trajectories estimated by the original LIBVISO2 and LIBVISO2-SURF-SIFT, vs the Ground Truth

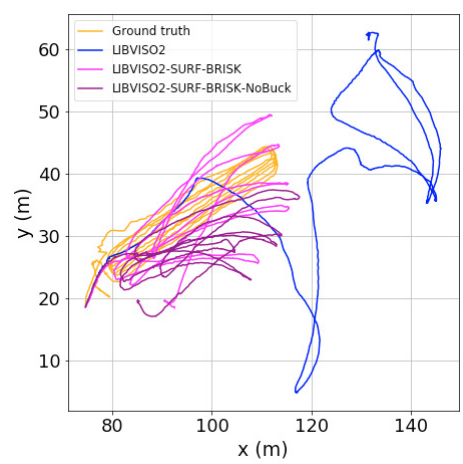

Fig. 6. Trajectories estimated by the original LIBVISO2 and LIBVISO2-SURF-BRISK, vs the Ground Truth

appear for the LIBVISO2-SURF-SIFT, with and without bucketing, d) the lowest lineal velocity mean error is for LIBVISO2-SIFT-SIFT-NoBuck, e) the lowest yaw mean error is for LIBVISO2-SURF-FREAK, and f) the lowest

Table 2. Mean errors in position, orientation, linear and angular velocities.

Algorithm	Position (m)	Yaw (rad)	Lineal vel. (m/s)	Angular vel. (rad/s)
LIBVISO2	35.24	2.11	0.033	0.0074
LIBVISO2-SIFT-SIFT	11.12	0.51	0.033	0.0034
LIBVISO2-SIFT-SIFT-NoBuck	11.10	0.58	0.030	0.0035
LIBVISO2-SURF-SIFT	23.31	2.39	0.087	0.0079
LIBVISO2-SURF-SIFT-NoBuck	21.69	2.31	0.080	0.0074
LIBVISO2-SURF-BRISK	7.01	0.98	0.038	0.0038
LIBVISO2-SURF-BRISK-NoBuck	8.31	1.16	0.035	0.0037
LIBVISO2-SURF-FREAK	9.57	0.16	0.050	0.0042
LIBVISO2-SURF-FREAK-NoBuck	15.45	1.13	0.046	0.0042

Table 3. Assessment metrics

Algorithm	Running time (s)	Number of matches	Number of inliers
LIBVISO2	0.34	148.05	47.15
LIBVISO2-SIFT-SIFT	0.66	78.47	59.12
LIBVISO2-SIFT-SIFT-NoBuck	0.68	300.98	231.55
LIBVISO2-SURF-SIFT	0.96	34.47	21.30
LIBVISO2-SURF-SIFT-NoBuck	0.96	77.97	48.03
LIBVISO2-SURF-BRISK	0.37	53.32	37.89
LIBVISO2-SURF-BRISK-NoBuck	0.39	167.76	118.71
LIBVISO2-SURF-FREAK	0.39	37.95	28.74
LIBVISO2-SURF-FREAK-NoBuck	0.40	107.87	80.52

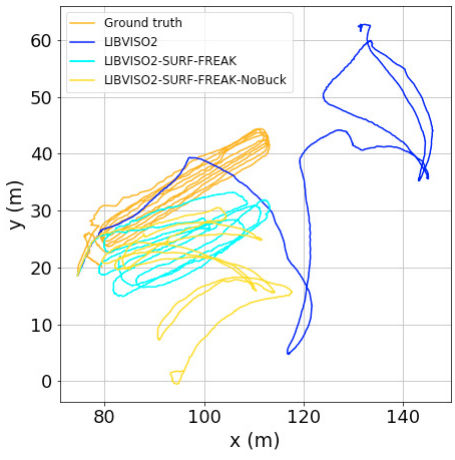

Fig. 7. Trajectories estimated by the original LIBVISO2, LIBVISO2-SURF-FREAK, vs the Ground Truth

accumulated drift is for the version LIBVISO2-SURF-BRISK, with and without bucketing, although the errors in lineal velocity and yaw are higher than the same errors for LIBVISO2-SIFT-SIFT.

According to Table 3: a) LIBVISO2, LIBVISO2-SURF-BRISK, with and without bucketing, and LIBVISO2-SURF-FREAK, with and without bucketing, are the versions with the shortest execution time, b) versions LIBVISO2-SURF-SIFT, with and without bucketing, are

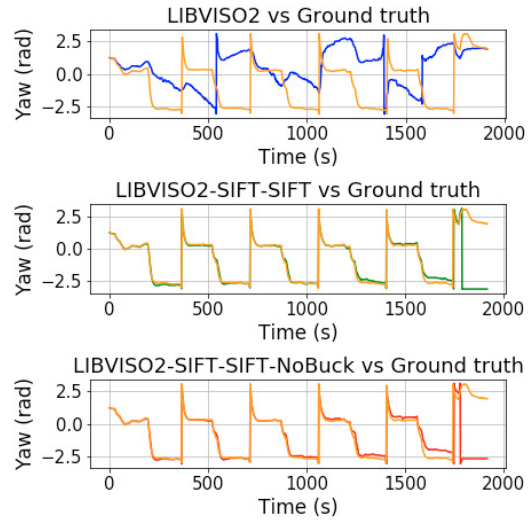

Fig. 8. Yaw evolution of original LIBVISO2 and LIBVISO2-SIFT-SIFT vs the Ground truth.

always the slowest, and c) LIBVISO2-SIFT-SIFT, with and without bucketing, have an intermediate runtime.

The time dedicated to estimate the camera displacement is perfectly acceptable for online applications, if the frame rate is configured to 2 fps and the AUV speed 0.25 m/s. The version with a maximum number of matches and inliers is LIBVISO2-SIFT-SIFT-NoBuck and the one with a minimum is LIBVISO2-SURF-SIFT.

From all these data, and taking into account that: 1) our interest is focused on getting a good performance in terms of estimating reliable camera displacements underwater rather than reducing the running time to the most, 2) the lineal velocity is one of the crucial elements to be integrated in the AUV navigation module, and 3) SIFT shows to offer the best performance for detection and description, the version that presents the best trade off between feature tracking robustness, trajectory accumulated drift and errors in lineal and angular velocities would be LIBVISO2-SIFT-SIFT-NoBuck, improving clearly the performance of the original LIBVISO2 in all terms. The bucketing procedure normally reduces the number of features, and, in this type of environments where stable and trackable features tend to disappear quickly, avoiding additional issues that can decrease the detection of successive image salient points is the best choice.

This new odometer has not been implemented to be used as an isolated source of localization or *localization data per se*, given the usual drifts provided by visual odometers

underwater, but to complement the navigation module that integrates all the other data given by the other instruments, in case one of them could run out. The influence of the odometric velocities in the global navigation will be modulated by its associated covariance, with a higher reliability as it decreases. In order to guarantee a reliable and robust localization underwater, visual SLAM would be the definitive solution, but this odometer can be a good base for the initial motion estimates.

Figure 8 shows the evolution of the vehicle orientation for the version LIBVISO2-SIFT-SIFT with and without bucketing, fitting very well with the *Ground Truth* estimated yaw.

5. CONCLUSION

This paper describes the changes made on the visual odometer LIBVISO2 to enhance its performance in underwater scenarios. Improvements were focused on strengthening the circular visual feature matching process that involves current and previous stereo pairs, and in the filtering of outliers. In the context of our ongoing projects, increasing the performance is preferable than reducing computation speed. Experiments performed with video sequences taken by the robot while doing lawn-mower shaped trajectories in the Balearic coast manifest the improvement of the new visual odometry estimates, in terms of pose errors and accumulated drift. A combination of SIFT detector with SIFT descriptor without bucketing is the configuration with the best trade off between pose and velocity mean errors and visual tracking performance, with respect to the original LIBVISO2 and the rest of versions implemented and tested.

Two different lines of work are being currently developed: first, integrating and running LIBVISO2-SIFT-SIFT-NoBuck as one of the fundamental blocks of the AUV navigation architecture, checking its influence online, during the missions, and, second, generating a more efficient feature detector for this type of environments using *Deep Learnig* and then coupling it to LIBVISO2-SIFT-SIFT.

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